ModelIt!

Teacher's Guide

The Disappearing Act

Ice Melts, But the Mass Never Leaves

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-PS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 5-PS1-2

Measure and graph quantities to provide evidence that the total weight of matter is conserved regardless of the type of change.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students investigate whether mass changes during state changes.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter** — Matter is conserved in all physical and chemical changes.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students measure and compare mass before and after changes.

CAST Testing Alignment

MULTIPLE CHOICE:

A student puts 100g of ice in a closed container. After the ice melts, the total mass will be:

CONSTRUCTED RESPONSE:

A candle starts at 50g. After burning for an hour, only 30g remains. Using conservation of matter, explain where the missing 20g went.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The law of conservation of matter states that matter cannot be created or destroyed, only transformed. This is one of the most fundamental principles in science.

Conservation Law

Antoine Lavoisier established the law of conservation of mass in 1789. He carefully measured chemical reactions and found that total mass always remained constant.

State Changes

When ice melts, water molecules don't disappear - they just move faster and spread apart. The same number of molecules exists in the liquid as in the solid.

Combustion

Burning appears to destroy matter, but actually converts solid carbon into CO2 gas and water vapor. If you capture the gases, total mass equals starting mass.

Digestion

When you digest food, the matter becomes part of your body, is exhaled as CO2, or is eliminated as waste. Nothing is destroyed.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify temperature, starting mass, ice amount, and liquid water amount as components.

E — Establish Relationships
Students establish that ice and liquid have a NEGATIVE relationship - they trade mass between them.

V — Visualize & Model
Students build the state change model with the key constraint of constant total mass.

E — Evaluate Outcomes
Students run heating and cooling scenarios, observing that Starting Mass never changes.

R — Revise & Extend
Students add water vapor as a third state to model complete water cycle.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"The ice 'disappeared' - but did it really?"*

Watch For

Set up demonstration if possible.

Slide 2: Learning Objectives

2 min

Say: *"Today we prove that mass is conserved."*

Watch For

Define conservation.

Slide 3: Big Question

2 min

Say: *"Where does the mass GO when ice melts?"*

Watch For

Take predictions.

Slide 4: LEVER Framework

2 min

Say: *"We'll model state changes."*

Watch For

Preview activities.

Slide 5: Activity 1: Components

8 min

Say: *"Starting Mass is our constant - it never changes."*

Watch For

Emphasize this key insight.

Slide 6: Activity 2: Connections

8 min

Say: *"Ice and Liquid are NEGATIVE because they share the same mass."*

Watch For

Demonstrate with a fixed amount of water.

Slide 7: Activity 3: Simulation

10 min

Say: *"Watch Starting Mass - it never moves!"*

Watch For

Students run multiple temperature scenarios.

Slide 8: Discoveries

5 min

Say: *"Conservation of matter is a LAW, not a suggestion."*

Watch For

Discuss implications.

Slide 9: STEM Challenge

10 min

Say: *"Apply conservation to solve a burning mystery."*

Watch For

Guide investigation of combustion.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Air Temperature, Starting Mass
- **INTERNAL:** Ice Amount, Liquid Water Amount

Relationships

- **Air Temperature to Ice Amount:** NEGATIVE (-) — Higher temperature means less ice (it melts).
- **Air Temperature to Liquid Water Amount:** POSITIVE (+) — Higher temperature means more liquid (from melting).
- **Ice Amount to Liquid Water Amount:** NEGATIVE (-) — They share the same mass - when one goes up, the other must go down.

Key Discoveries

- Mass is conserved - it doesn't disappear when ice melts
- Ice Amount and Liquid Water Amount are mirror images of each other
- Starting Mass stays completely flat regardless of temperature
- State changes are physical changes that conserve mass

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Ice weighs more than water because it's solid"	Same mass, different density. Ice actually takes up MORE space than the same mass of water.
"Matter disappears when it evaporates"	Water vapor is still matter - it's just invisible gas molecules. The mass is in the air.
"Burning destroys matter"	Combustion converts solid to gases (CO_2, H_2O). Capture the gases and total mass is conserved.